

Introduction

1. Data and Signal

Data	Signal
<i>Data is an entity that conveys meaning or information, interpreted based on agreed-upon rules between sender and receiver</i>	<i>A signal is the physical representation of data transmitted over a medium — electrical, optical, or electromagnetic.</i>
<i>Data can be analog (e.g., voice, temperature) or digital (e.g., binary values).</i>	<i>A signal can also be analog (a continuous wave) or digital (discrete voltage levels).</i>
<i>Data originates at the source before any transmission takes place.</i>	<i>A signal is what carries that data through the transmission medium.</i>
<i>Data is measured in units of information such as bits or bytes.</i>	<i>A signal is measured in physical units such as volts, hertz, or decibels.</i>
<i>Data cannot travel on its own — it must be encoded into a signal first.</i>	<i>A signal can travel through both guided media (cables) and unguided media (air).</i>
<i>An example of data is the number "65" or the letter "A".</i>	<i>An example of a signal is the voltage pattern on a wire that represents "65".</i>

2. Data Flow

Simplex

In simplex mode, data flows in only one direction — one device always sends and the other only receives. A keyboard sending input to a computer is a classic example.

Half-Duplex

In half-duplex mode, both devices can send and receive, but not at the same time — they must take turns. A walkie-talkie is the most common example.

Full-Duplex

In full-duplex mode, both devices can send and receive simultaneously without waiting for the other to finish. A phone call is the standard example.

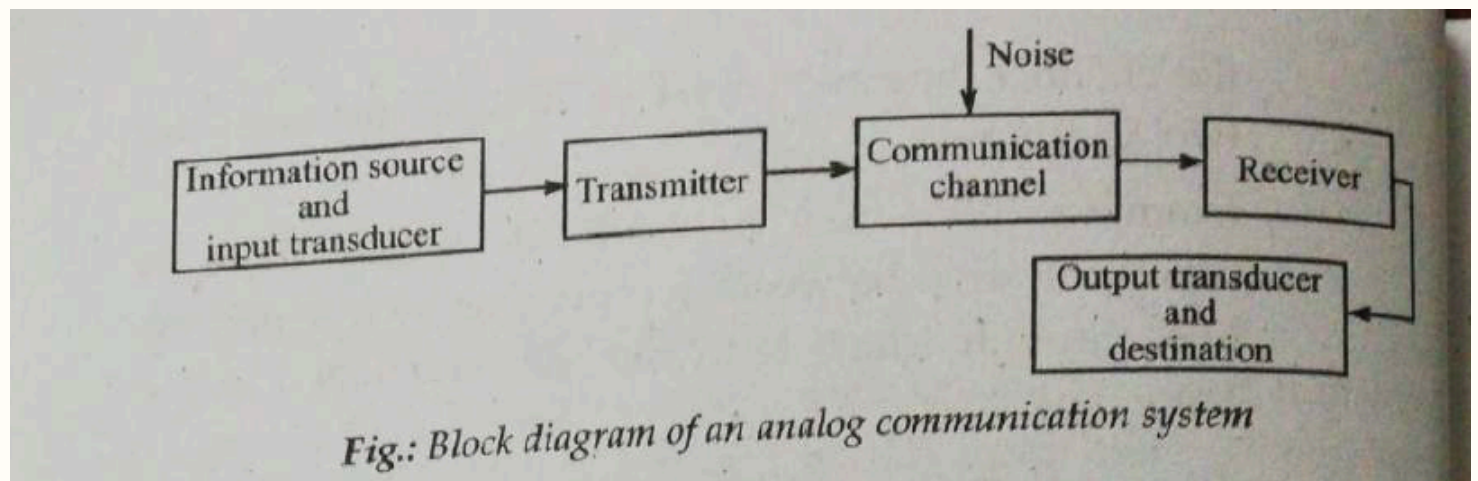
3. Communication System

1. Analog Communication System

An analog communication system transmits information using continuous signals that vary in amplitude or frequency. The input (such as voice) is converted to an electrical signal by a transducer, then modulated by the transmitter to make it suitable for channel transmission. At the receiving end, the signal is demodulated to recover the original message and converted back to its physical form.

Flow:

Source → Input Transducer → Transmitter → Channel → Receiver → Output Transducer → Destination

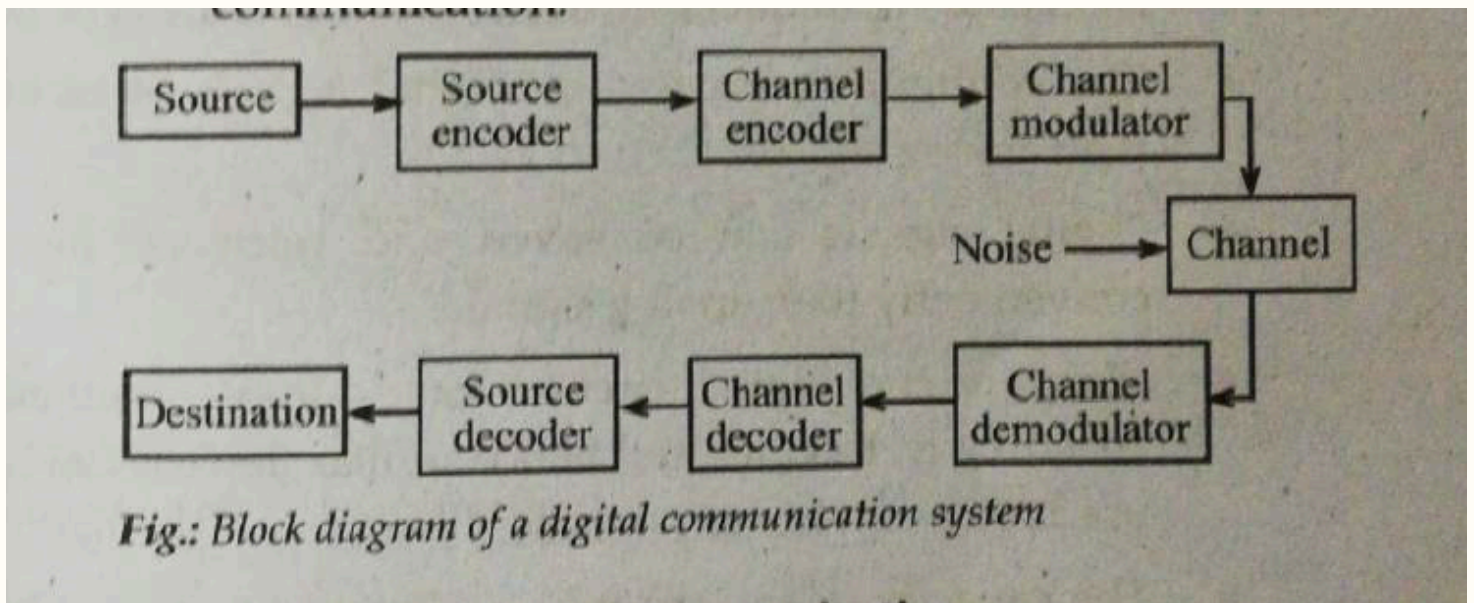


1. **Input Transducer:** Converts physical quantity → electrical signal (e.g., microphone).
2. **Transmitter:** Processes message into transmittable form; key operation is **modulation**.
3. **Channel:** Physical medium (wire, air, fiber) linking sender and receiver.
4. **Receiver:** Recovers message from received signal; key operation is **demodulation**.

5. **Output Transducer:** Converts electrical signal → physical output (e.g., speaker).

2. Digital Communication System

A digital communication system transmits information as discrete binary data (0s and 1s). The source data is first encoded to remove redundancy, then channel-coded to add error protection, and finally modulated for channel transmission. At the receiver, these operations are reversed — demodulation, error correction, and decoding — to faithfully recover the original data. Digital systems are preferred over analog because they offer better noise immunity, error correction, and security.



1. **Source:** Generates digital data (text, image, audio, etc.)
2. **Source Encoder:** Converts data to binary; removes redundancy (compression).
3. **Channel Encoder:** Adds error detection/correction bits.
4. **Modulator:** Converts bits into a signal suitable for the channel.
5. **Channel:** Transmission medium; signal may be corrupted by noise here.
6. **Demodulator:** Converts received signal back to binary.
7. **Channel Decoder:** Detects and corrects errors using added bits.
8. **Source Decoder:** Reconstructs original data from binary.
9. **Destination:** Final receiver of the information.

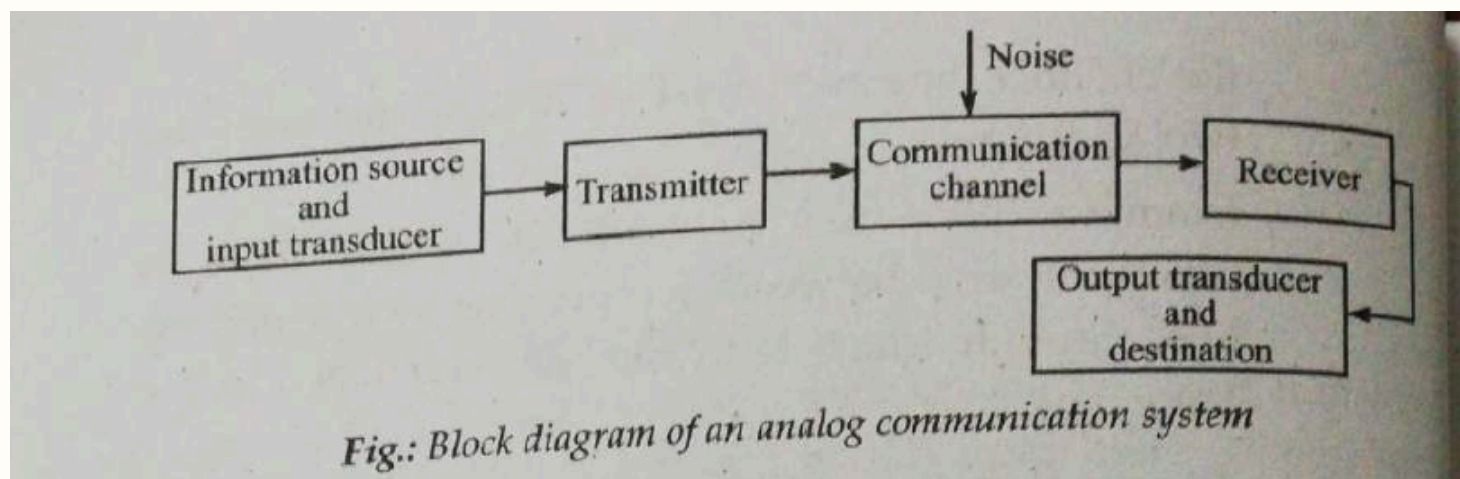
Flow:

Source → Source Encoder → Channel Encoder → Modulator → Channel →
Demodulator → Channel Decoder → Source Decoder → Destination

3. Analog vs Digital System

Feature	Analog	Digital
Signal Type	Continuous (sine wave)	Discrete (0s and 1s)
Noise Effect	Highly affected	Less affected
Transmission Quality	Degrades over distance	Maintained via regenerators
Error Detection	Difficult	Easy (error codes)
Security	Less secure	Highly secure (encryption)
Example	Radio, TV	Internet, mobile

4. Transmission Mode



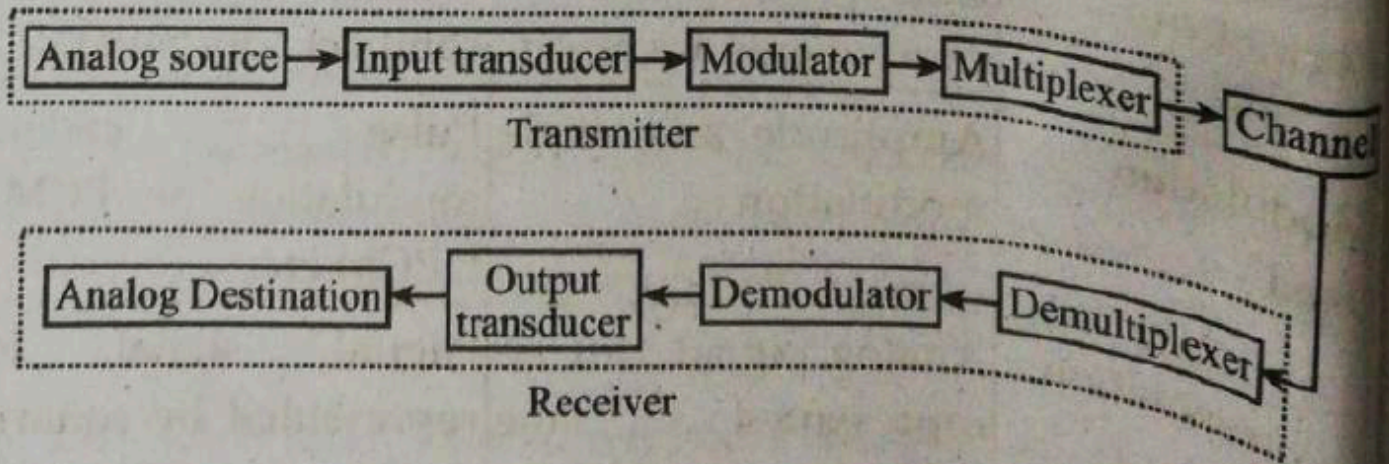


Fig.: Half-duplex analog communication system

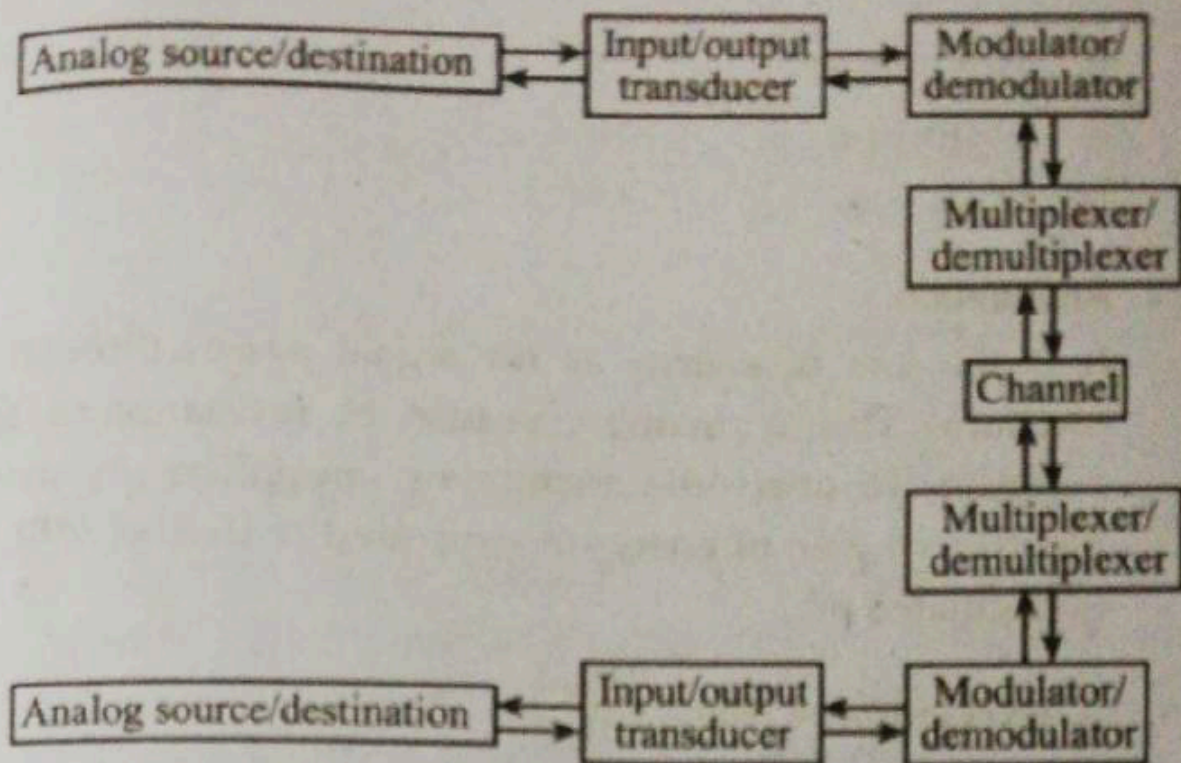


Fig.: Full-duplex analog communication system

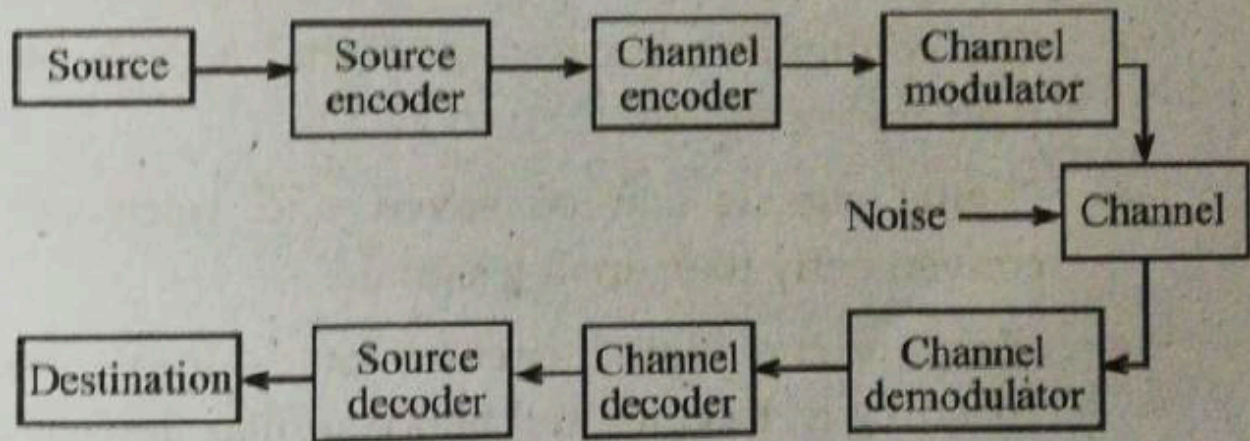


Fig.: Block diagram of a digital communication system

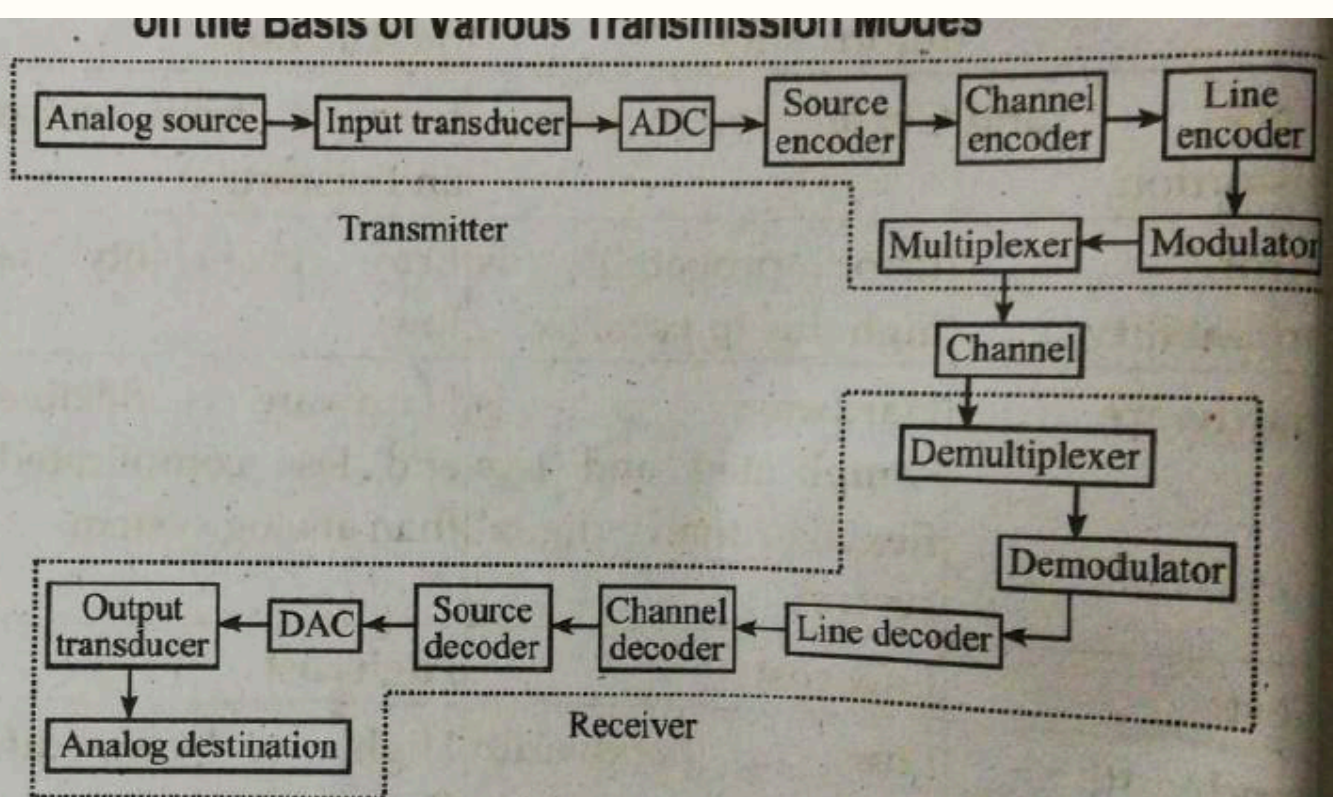
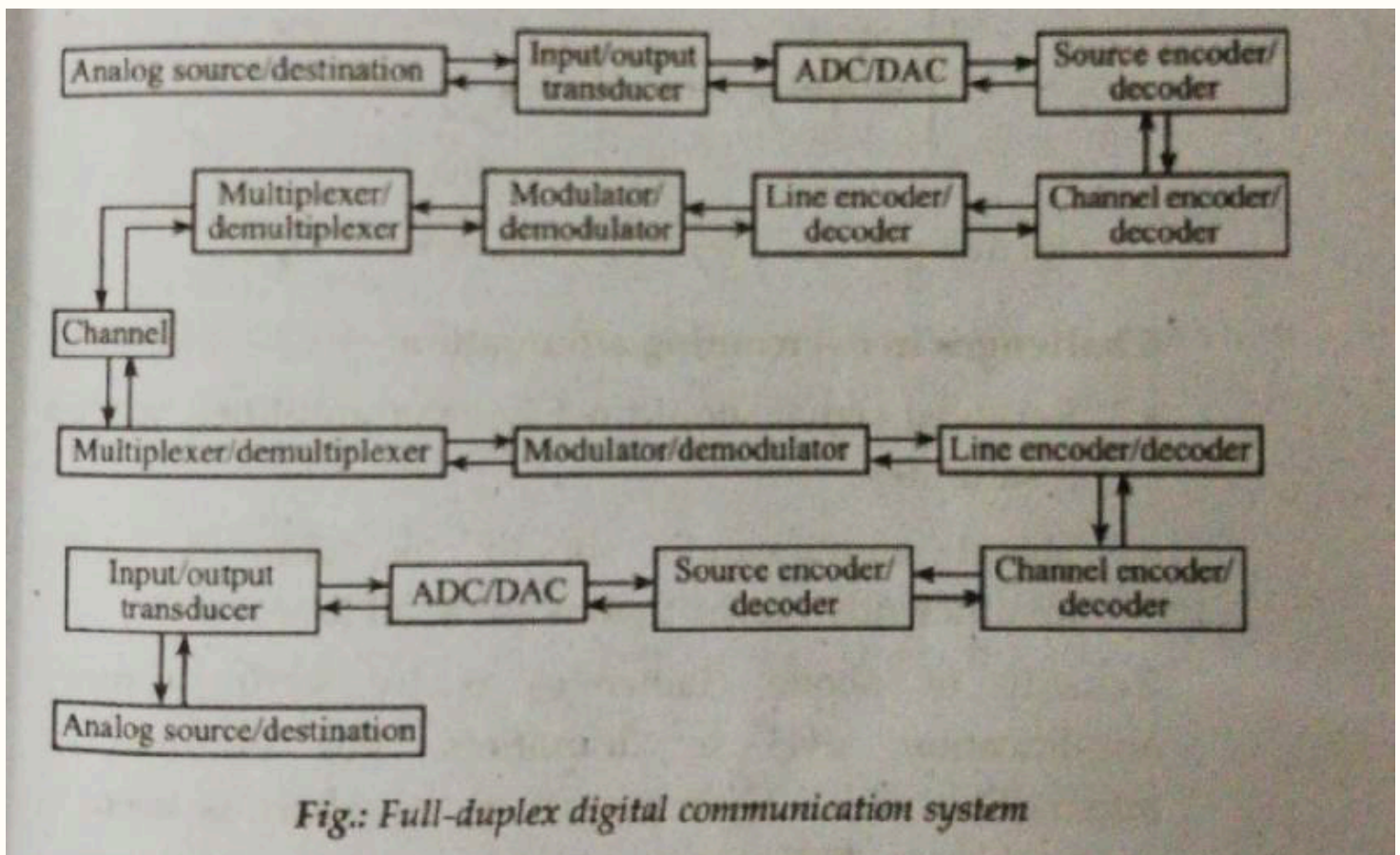


Fig.: Half-duplex digital communication system



5. Transmission Impairments

Transmission impairments are factors that degrade signal quality during transmission over a channel. The three main types are attenuation, distortion, and noise.

1. Attenuation:

Loss of signal strength as it travels through the medium over distance. The signal becomes weaker and may become unreadable. Fixed using **amplifiers** (analog) or **repeaters** (digital).

2. Distortion:

The shape or form of the signal changes during transmission. Types:

- **Amplitude distortion** — different frequencies amplified unequally.
- **Delay distortion** — different frequencies arrive at different times.
- **Phase distortion** — phase shift introduced in the signal.

Fixed using **equalizers**.

3. Noise:

Unwanted signals that mix with the original signal, corrupting it. Types:

- **Thermal noise** — caused by random electron movement due to heat; always present.
- **Intermodulation noise** — signals on different frequencies mix and produce unwanted signals.
- **Crosstalk** — signal from one channel leaks into another.
- **Impulse noise** — sudden, short-duration spikes caused by external events (e.g., lightning).

6. Performance Parameters of Communication Channel

1. Bandwidth (BW)

- Range of frequencies a channel can carry.

2. Throughput

- Actual amount of data successfully transmitted per unit time.

3. Latency (Delay)

- Total time for first bit to travel from sender to receiver.

4. Jitter

- Variation in packet arrival time.
- Caused by variable queuing/propagation delays.
- Critical for real-time applications (VoIP, video streaming).
- High jitter → poor audio/video quality.

5. Bit Error Rate (BER)

- Ratio of incorrectly received bits to total transmitted bits.
- Lower BER → better channel quality.
- Affected by noise, attenuation, interference.

7. Thermal Noise Power Density

Thermal noise (Johnson noise) is caused by random motion of electrons due to heat.

Formula:

$$N_0 = k \cdot T \quad (\text{W/Hz})$$

Where:

- k = Boltzmann's constant = $1.38 \times 10^{-23} \text{ J/K}$
- T = Temperature in Kelvin = $^{\circ}\text{C} + 273$

Total noise power over bandwidth B :

$$N = k \cdot T \cdot B \quad (\text{Watts})$$

Solved Example (from 2071 set):

Calculate thermal noise power density at $T = 17^{\circ}\text{C}$

$$T = 17 + 273 = 290 \text{ K}$$

$$N_0 = 1.38 \times 10^{-23} \times 290$$

$$N_0 = 4.002 \times 10^{-21} \text{ W/Hz}$$

$$N_0 \approx 4 \times 10^{-21} \text{ W/Hz}$$